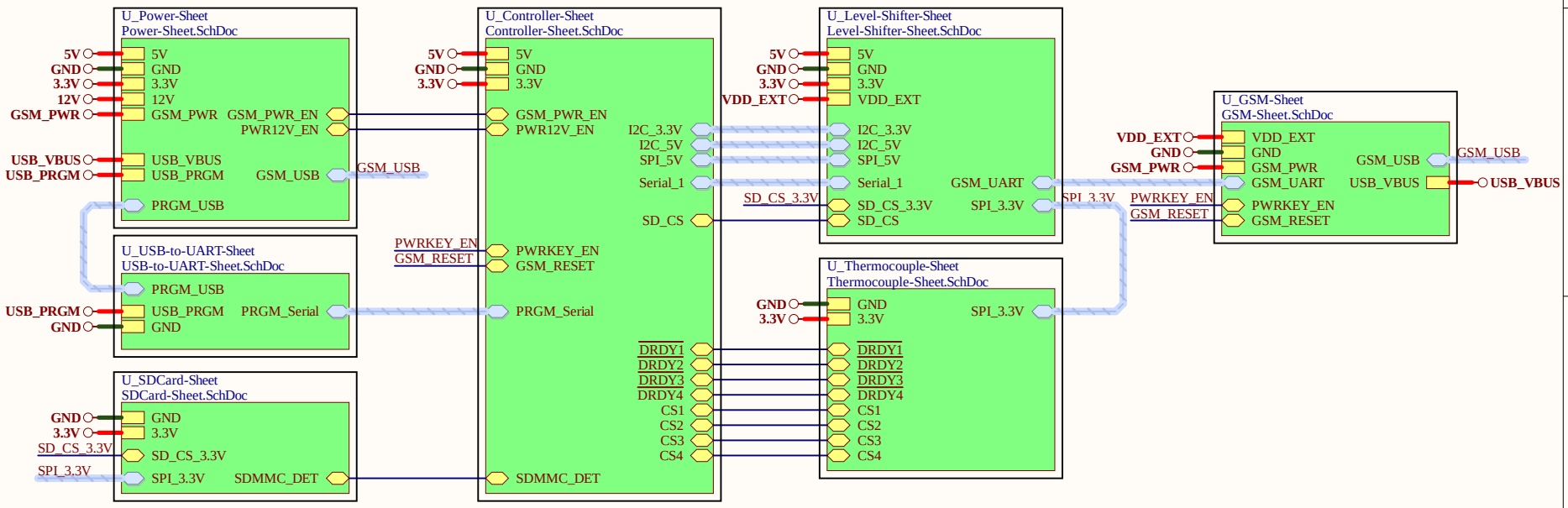
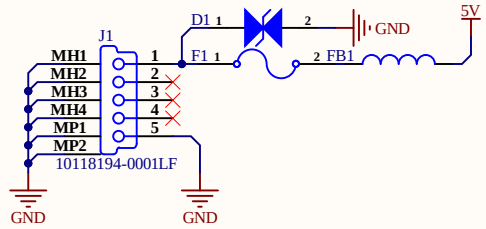


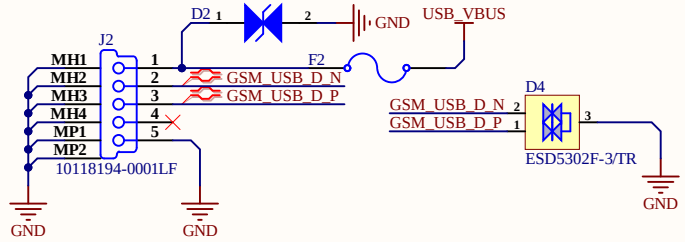
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File: Mega2560_Custom.PrjPcb	
Date: 1/23/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 8

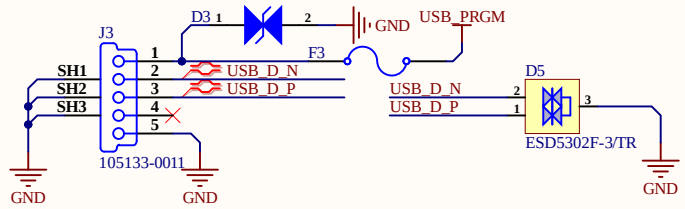
USB to Power Board



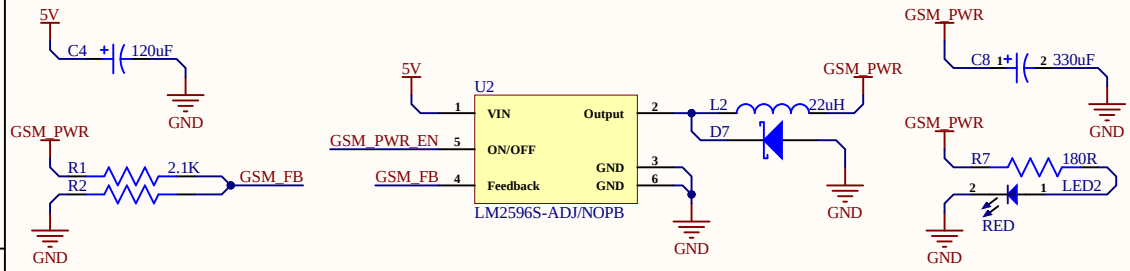
USB for GSM Updates



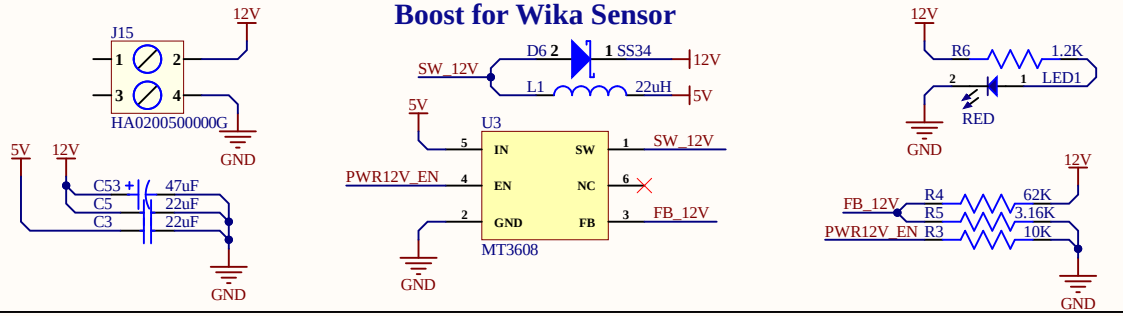
USB for Programming Mega2560



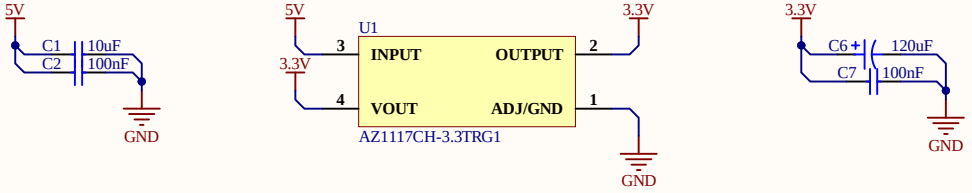
Buck for GSM Power



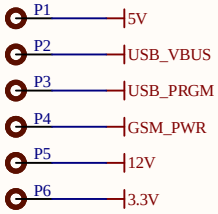
Boost for Wika Sensor

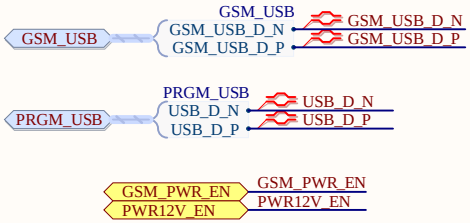


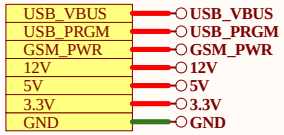
LDO 3.3V



Test Points



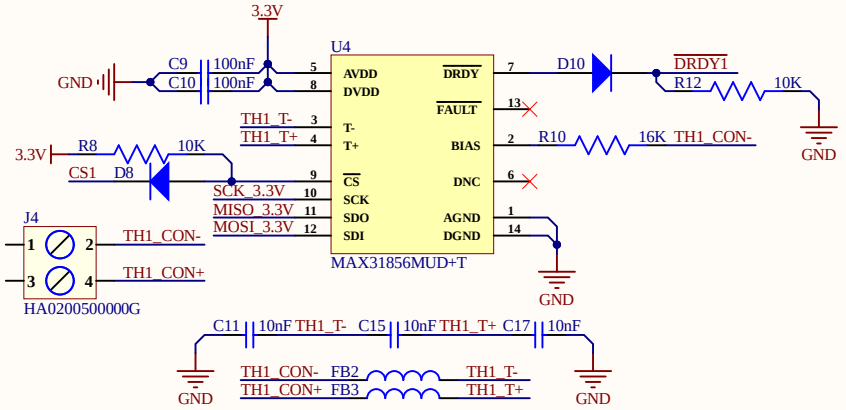




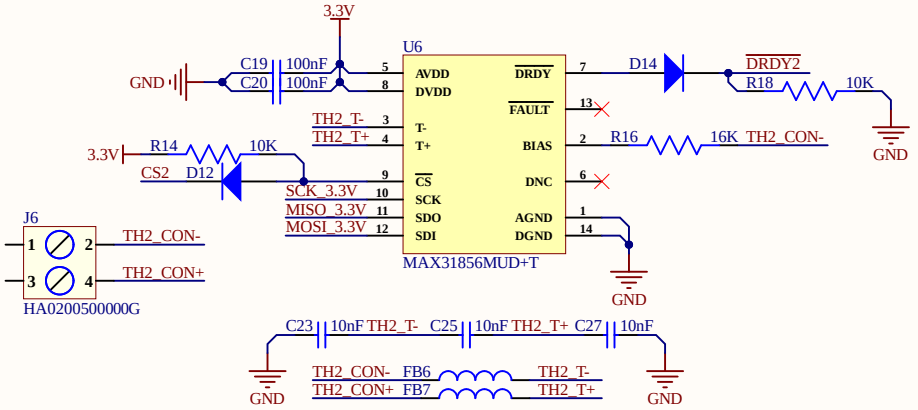
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Date: 1/23/2023	Page Num: 2
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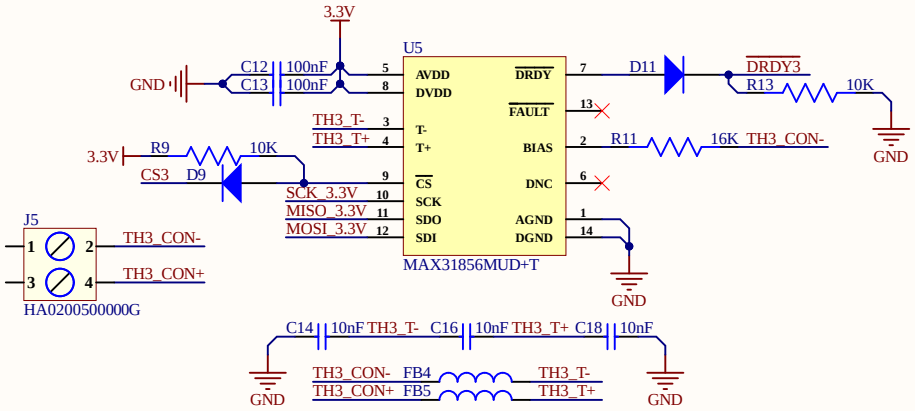
Thermocouple Input - 1



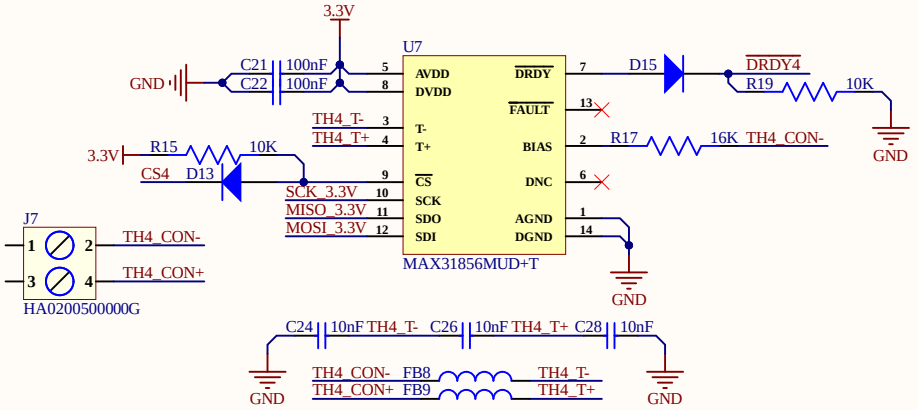
Thermocouple Input - 2



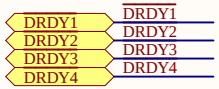
Thermocouple Input - 3

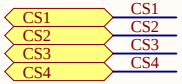


Thermocouple Input - 4







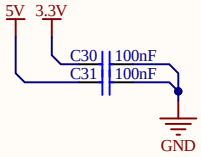


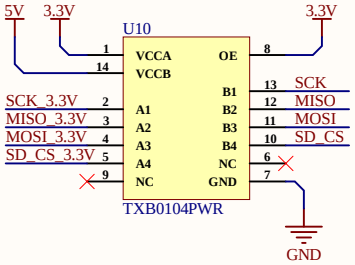


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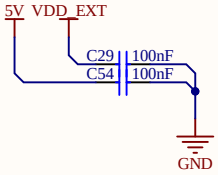
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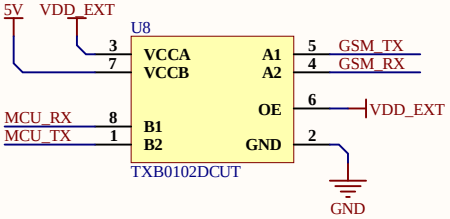
Level Shifter for SPI



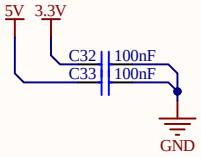


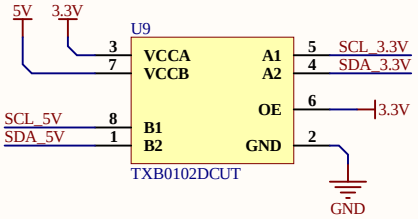
Level Shifter for Serial

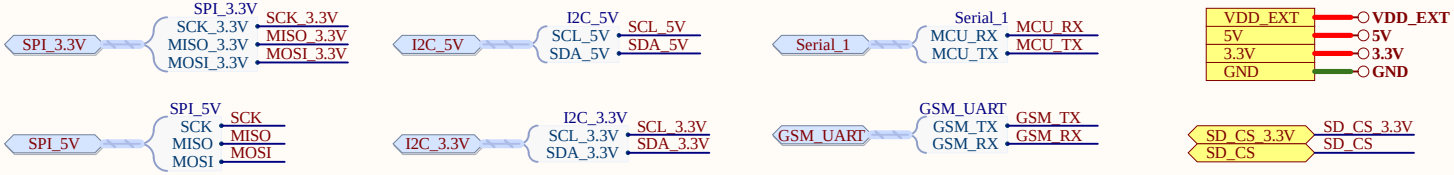




Level Shifter for I2C

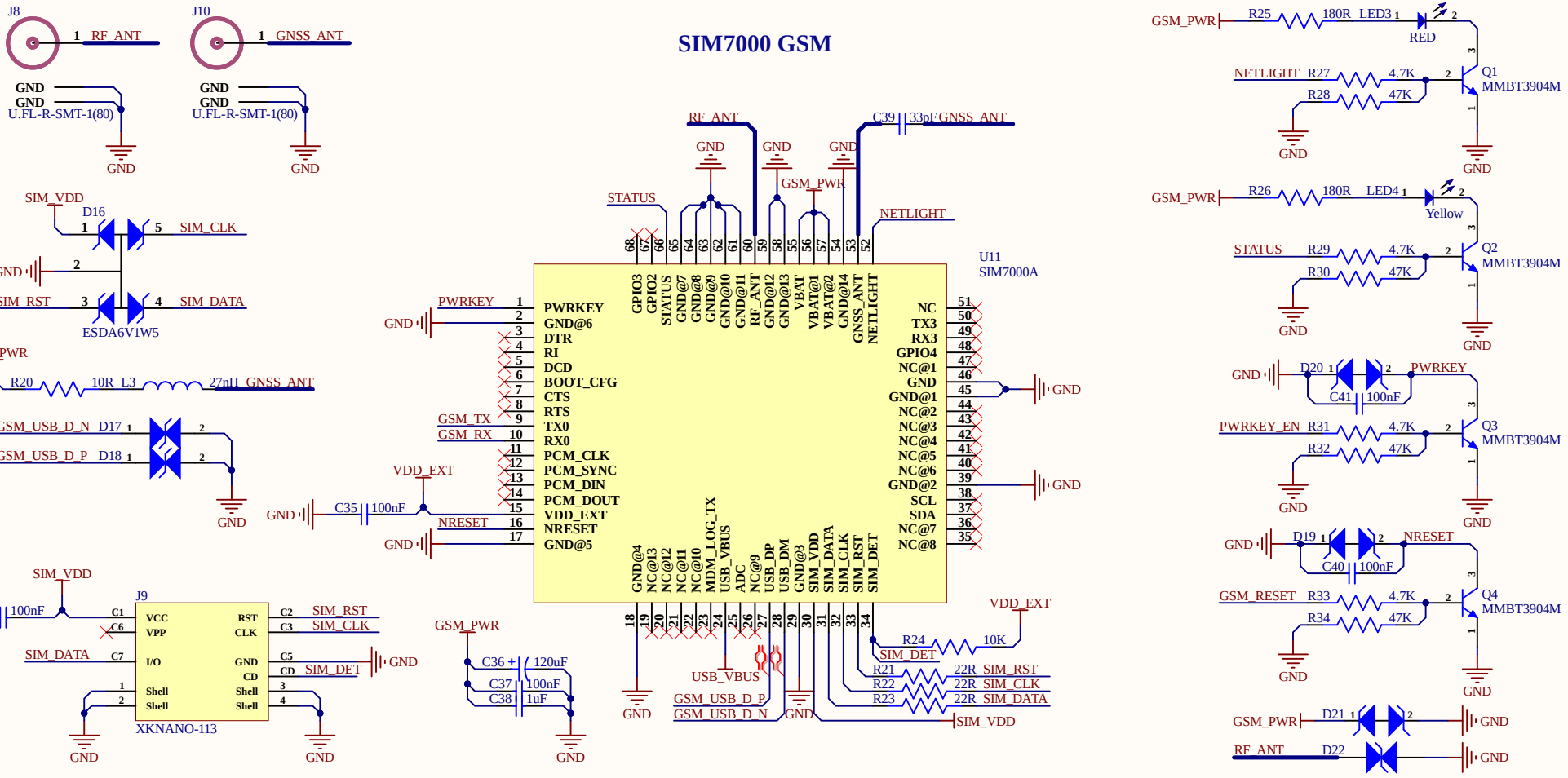






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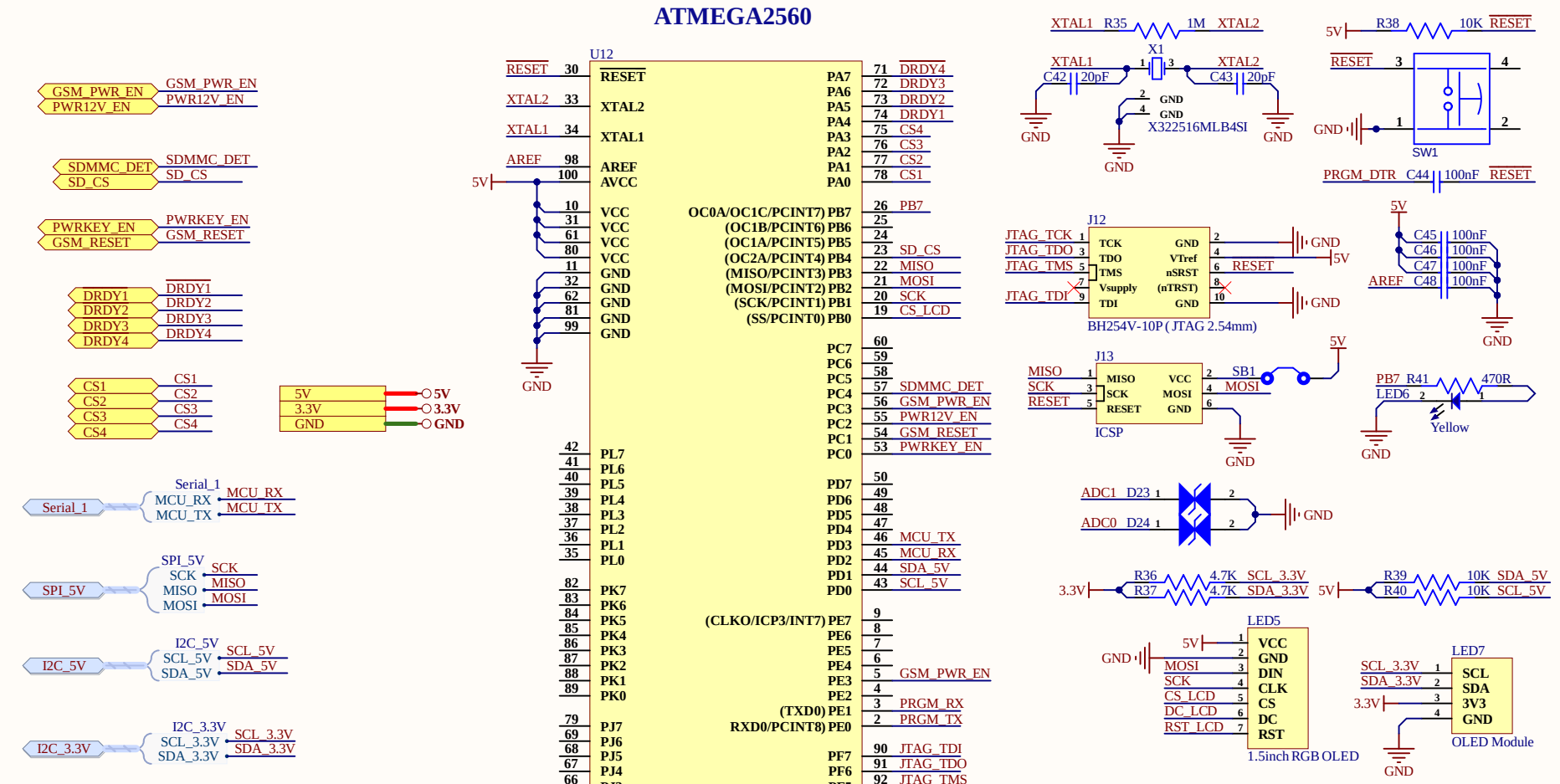
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Date: 1/23/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 8



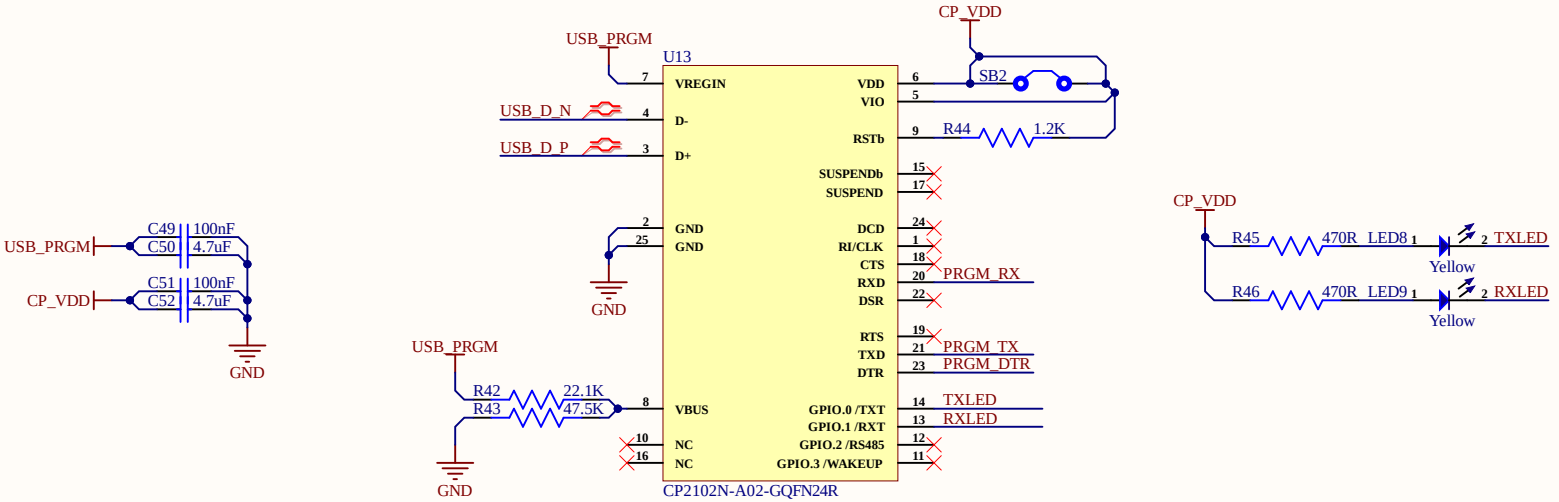
GSM_UART
GSM_TX
GSM_RX
GSM_USB
GSM_USB_D_N
GSM_USB_D_P
PWRKEY_EN
GSM_RESET
USB_VBUS
GSM_PWR
VDD_EXT
GND

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Title: GSM-Sheet.SchDoc	
File: Mega2560_Custom.PrjPcb	
Date: 1/23/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 8


ATMEGA2560
U12
ATMEGA2560-16AU
GSM_PWR_EN
PWR12V_EN
SDMMC_DET
SD_CS
PWRKEY_EN
GSM_RESET
DRDY1
DRDY2
DRDY3
DRDY4
CS1
CS2
CS3
CS4
5V
3.3V
GND
Serial_1
MCU_RX
MCU_TX
SPI_5V
SCK
MISO
MOSI
I2C_5V
SCL_5V
SDA_5V
I2C_3.3V
SCL_3.3V
SDA_3.3V
PRGM_Serial
PRGM_RX
PRGM_TX
PRGM_DTR
Analog Inputs
J11
ADC1
ADC0
HA0200500000G
XTAL1
XTAL2
R35
1M
X1
C42
C43
20pF
X322516MLB4SI
R38
10K
SW1
PRGM_DTR
C44
100nF
J12
JTAG_TCK
JTAG_TDO
JTAG_TMS
JTAG_TDI
BH254V-10P (JTAG 2.54mm)
C45
C46
C47
C48
AREF
J13
ICSP
PB7
R41
470R
LED6
Yellow
ADC1
D23
ADC0
D24
R36
R37
4.7K
R39
R40
10K
LED5
1.5inch RGB OLED
LED7
OLED Module
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Title: Controller-Sheet.SchDoc
File: Mega2560_Custom.PrjPcb
Date: 1/23/2023
Page Num: 6
Drawn By: Huzaifa Ali
Page Total: 8

USB-to-UART for Programming





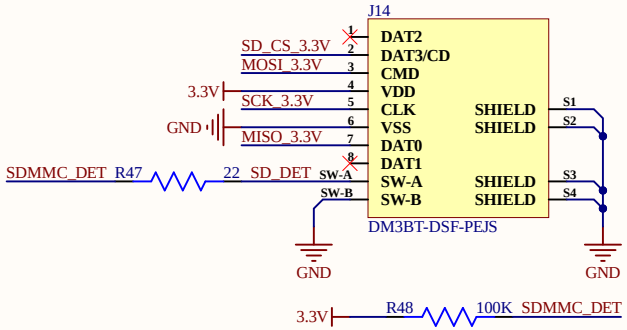


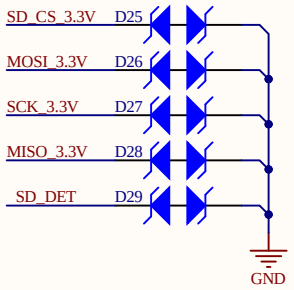


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Title: USB-to-UART-Sheet.SchDoc	
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SD Card Interface





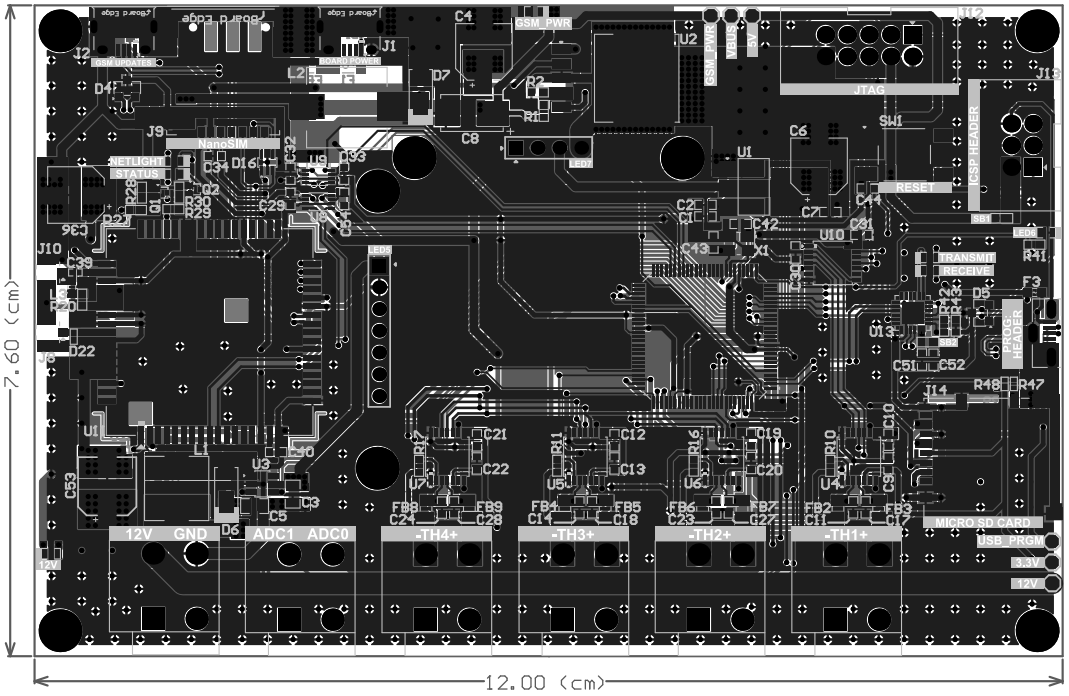






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Title: SDCard-Sheet.SchDoc	
File: Mega2560_Custom.PrjPcb	
Date: 1/23/2023	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 8



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	3.00mil	3.5	
1	Top Layer		10.00mil		
	Dielectric 1	FR-4	37.00mil	4.8	
2	Bottom Layer		10.00mil		
	Bottom Solder	Solder Resist	3.00mil	3.5	
	Bottom Overlay				